Project Description: Homological Algebra Beyond Projective Space

1. The Geometry of Syzygies on Stacks

1.1 N_p-Conditions for Stacky Curves

Research Goal 1.1. *Understand the syzygies of stacky curves.*

Research Problem 1.2. *Let $\mathcal{X}$ be a stacky curve of genus 0 and 1. If $\mathcal{L}$ is a line bundle on $\mathcal{X}$ and $\deg(\mathcal{X}) \gg 0$ what N_p conditions does $R(\mathcal{X}; \mathcal{L})$ satisfy?*

1.2 Green's Conjecture for Canonical Stacky Curves

Research Goal 1.3. *Find and prove an appropriate generalization of Green's conjecture describing the vanishing of the syzygies of the canonical rings of stacky curves.*

Theorem 1.4. [?] **Theorem 8.4.1 Let $\mathcal{X}$ be stacky curve whose coarse space has genus ≥ 1 and let e be the maximum order of the stabilizer groups of the stacky points; then the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ is generated by elements in degree at most $3e$ with relations in degree at most $6e$.*

Research Problem 1.5. *Can Theorem 1.4 be refined by considering the existence of certain stacky or weighted linear series on $\mathcal{X}$?*

Research Problem 1.6. *If $\mathcal{X}$ is a general stacky curve is the stacky Brill-Noether locus $\mathcal{W}_d^r(\mathcal{X})$ of expected dimension, smooth away from $\mathcal{W}_d^{r+1}(\mathcal{X})$, and irreducible when of positive dimensional?*

Research Problem 1.7. *Let $\mathcal{X}$ be a stacky whose coarse space has genus g. For what k does the canonical ring $R(\mathcal{X}; K_{\mathcal{K}})$ satisfy begin k-Koszul in the sense above.*

Research Problem 1.8. *Compute the minimal graded free resolution of the canonical ring $R(\mathcal{X}, K_{\mathcal{X}})$ for all stacky curves $\mathcal{X}$ where:*

- *the coarse space of $\mathcal{X}$ has genus ≤ 4,*
- *there are at most 4 stacky points, and*
- *the stabilizers of the stacky points have order at most 5.*

1.3 Asymptotic Syzygies of Weighted Projective Stacks

Research Problem 1.9. *Let $\mathcal{X}$ be a smooth DM stack, $\dim \mathcal{X} \geq 2$, and fix an index $1 \leq q \leq \dim \mathcal{X}$. If $(L_d)_{d \in \mathbb{N}}$ is a sequence of very ample line bundles such that $L_{d+1} - L_d$ is constant and ample compute the limit (if it exists):*

$$\lim_{d\to\infty} \rho_q(\mathcal{X}; L_d) = \lim_{d\to\infty} \frac{\#\left\{p \in \mathbb{N} \;\middle|\; \beta_{p,p+q}(X, L_d) \neq 0\right\}}{n_d}.$$

Research Problem 1.10. *Fix a positive integer $a > 1$. For what values of p does the weighted projective stacky plane $\mathcal{P}(1,1,1,a)$ embedded by $\mathcal{O}(d)$ satitify N_p.*

1.4 Computations with Toric Stacks

Research Problem 1.11. *Develop a robust software package for Macaualy2 allowing for computations with toric stacks.*

Research Problem 1.12. *Compute the graded Betti tables of $\mathcal{P}(1,1,1,a)$ embedded by $\mathcal{O}(d)$ for all $1 < a \leq 15$ and all $2 \leq d \leq 6$ by adapting the high-performance computing approaches developed in* ♠♠♠ Juliette: [cite]

2. Geometric & Algebraic Measures of Complexity on Toric Varieties

2.1 Asymptotic Regularity of Powers of Ideals and Ideal Sheaves

Research Goal 2.1. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring. Understand the asymptotic behavior of $\operatorname{reg}(I^p)$ and $\operatorname{reg}(\mathcal{I}^p)$ where $I \subset S$ is an $\mathbb{Z}^r$-graded ideal and $\mathcal{I} \subset \mathcal{O}_X$ is an ideal sheaf.*

Conjecture 2.2. *Let X be a smooth projective toric variety and $\mathcal{I} \subset \mathcal{O}_X$ an ideal sheaf. With the notation as above:*

$$\lim_{p\to\infty} \frac{\operatorname{reg}(\mathcal{I}^p)_{\mathbb{R}}}{p} = \left\{ L \in\in \operatorname{Pic}(X)_{\mathbb{R}} \;\middle|\; \pi^*(L) - E \text{ is nef on } \tilde{X} \right\}.$$

Research Problem 2.3. *Prove Conjecture 2.2.*

Research Problem 2.4. *Is the Seshadri regions defined above be determined by the Newton-Okunkov body of some flag on X.*

Research Problem 2.5. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring. If $I \subset S$ is a $\mathbb{Z}^r$-graded ideal bound the number of minimal generators of $\operatorname{reg}(I^p)$ as a function of p in terms of invariants of I?*

Research Problem 2.6. *Let X be a smooth projective toric variety and $Z \subset X$ a smooth subvariety. Can one bound the multigraded regularity of I_Z in terms of the codimension of $Z \subset X$ and the degrees of minimal generators of I_Z?*

Research Problem 2.7. *Compute the multigraded regularity of torus invariant subvarieties of X in terms of the combinatorics of the cone defining them.*

2.2 A Generalized Eisenbud–Goto Theorem for Toric Varieties

Research Problem 2.8. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring and B its irelevant ideal. Let M be a $\operatorname{Pic}(X)$-graded S-module. Find the correction generalization of d-quasi-linear resolution such that a $\operatorname{Pic}(X)$-graded S-module M is d-regular if and only if the truncation $M_{\geq d}$ has a d-quasilinear resolution.*

Conjecture 2.9. *Let S be the $\mathbb{Z}^2$-graded Cox ring of the Hirzebruch surface H_t. If M is a $\mathbb{Z}^2$-graded S-module then $M_{\geq d}$ has a d-linear resolution if and only if $d + \operatorname{reg}(S) \subset \operatorname{reg}(M)$.*

Research Problem 2.10. *Prove Conjecture 2.9*

Research Problem 2.11. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring and B its irrelevant ideal. Let M be a $\operatorname{Pic}(X)$-graded S-module. Find a characterization for when the truncation $M_{\geq d}$ for $d \in \operatorname{Pic}(X)$ has a linear resolution in terms of vanishing of the local cohomology groups $H^i_B(M)_t$.*

Research Problem 2.12. *Find an algorithm to compute a minimal generating set for $\operatorname{reg}(M)$.*

Research Problem 2.13. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring and B its irrelevant ideal. Let M be a $\operatorname{Pic}(X)$-graded S-module. Does the set of (multigraded) Betti numbers of all virtual resolutions of M determine the multigraded regularity of M?*

2.3 Uniform Homological Algebra via D_{Cox}

Research Problem 2.14. *Develop a theory of Castelunovo–Mumford regularity on the category D_{Cox} the restrict to the classical notion of regularity X_i.*

Research Problem 2.15. *Let $I \subset S$ be a $\mathbb{Z}^r$-graded homogenous ideal. Let X and X' be toric varieties corresponding to two chambers $\Gamma, \Gamma' \subset \Sigma_{GKZ}$ that share a wall, with $B, B' \subset S$ the respective irrelevant ideals. Prove there exists a "small" subset $F \subset \mathbb{Z}^r$ depending only on X and X' such that:*

$$\operatorname{reg}_X\left((J :_S B^\infty)\right) + F = \operatorname{reg}_{X'}\left((J :_S B'^\infty)\right) + F.$$

Research Problem 2.16. *With the set up as above prove the minimal graded free S-resolutions of $(J :_S B^\infty)$ and $(J :_S B'^\infty)$ agree in small degree.*

2.4 The Free Resolution of $\overline{\mathcal{M}}_{0,n}$

Research Problem 2.17. *Compute the multigraded Castelunovo–Mumford Regularity of $\overline{\mathcal{M}}_{0,n+3} \subset \mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^n$.*

Research Problem 2.18. *Compute the minimal graded free resolution of the defining ideal I of $\overline{\mathcal{M}}_{0,n+3} \subset \mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^n$.*

Research Goal 2.19. *Describe the minimal graded free resolutions or multigraded regularitiy of wonderful varieties.*